

Supplementary information

Characterisation and Quantification of Data Centre Flexibility for Power System Support

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This document specifies the inputs, formulation and numerical interpretation of the annual and event results. Tables of inputs and outputs accompany the source package as machine-readable CSV files. All equations describe the model used for the retained results; no revised dispatch has been generated for this manuscript.

S1 Notation, parameters and initial conditions

Electrical powers are in kW, thermal powers in kW-th, stored energies in kWh or kWh-th, temperatures in degrees Celsius, and computational work in aggregate CPU-hours. Thermal heat capacities and conductances below use kJ/K and kW/K. The implementation uses W and J for several cooling quantities; the equations here convert these consistently to kW and kJ. A rate of one denotes the entire modelled computing capacity.

Table S1: Principal notation.

Symbol	Meaning
$t, \Delta t, \Delta t_s$	Interval start; interval length 0.25 h or 900 s. Boundary $t + 1$ follows interval t .
$d, C_d, \mathcal{H}_d$	Local date, committed core intervals, core plus twelve look-ahead intervals.
c, r_c, ℓ_c, a_c	Cohort, arrival, final eligible execution-interval start, original CPU-hour requirement.
$A_{c,d}, R_{c,H}$	Work available to process at horizon opening (or arrival within it), and remaining work at boundary H .
$u_t^{\text{inflex}}, u_t^{\text{flex,arr}}, u_{c,t}, u_t$	Inflexible and flexible arrival rates, scheduled cohort rate, total executed utilisation.
$\alpha_{t,k}, D_k$	Fraction of flexible arrival assigned to class k , and its maximum shift in interval starts.
$P_t^{\text{IT}}, P_t^{\text{grid}}, P^{\text{aux}}$	IT power, site import, fixed auxiliary power.
$P_t^{\text{U,ch}}, P_t^{\text{U,dis}}, E_t^{\text{U}}, z_t^{\text{U}}$	UPS charge/discharge power, stored energy and charge-mode binary.
$P_t^{\text{C}}, P_t^{\text{T,ch}}$	Chiller electricity for direct CRAC cooling and TES charging.
$Q_t^{\text{T,ch}}, Q_t^{\text{T,dis}}, E_t^{\text{T}}, z_t^{\text{T}}$	TES thermal charge/discharge, stored cooling energy and charge-mode binary.
Q_t^{cool}	Gross cooling supply from direct chiller and TES paths.
$T^{\text{IT}}, T^{\text{R}}, T^{\text{CA}}, T^{\text{HA}}, T^{\text{in}}$	IT, rack, cold-aisle and hot-aisle state temperatures; interval inlet temperature.
$C_j, G_{\text{conv}}, G_{\text{cold}}, \dot{m}, c_p, \kappa$	Thermal capacities, IT-rack and cold-aisle conductances, air flow, specific heat and effective-airflow factor.

Symbol	Meaning
$\pi_t, C^{\text{ref}}, C^{\text{op}}$	Signed reference price; annual comparator and coordinated costs.
$t_0, \Delta P, \tau, H$	Event start, signed import deviation, event duration and recovery boundary.
Superscripts ref, op	Workload-at-arrival comparator and undisturbed coordinated operation.

Table S2: Thermal constants from the implemented parameter definitions (rounded here to six decimal places).

Parameter	Value	Unit
C_{IT}	17880.000000	kJ/K
C_{R}	18019.516902	kJ/K
C_{CA}	2332.644000	kJ/K
C_{HA}	1166.322000	kJ/K
G_{conv}	108.991826	kW/K
G_{cold}	4.483672	kW/K
c_p	1.005450	kJ/(kg K)
$\dot{m}$	100.000000	kg/s
κ	0.766300	—
T^{out}	22.000000	°C

The principal electrical and storage settings are: IT idle/rated power 166.7/1,000 kW; UPS energy 600 kWh, bounds 300–600 kWh, initial energy 600 kWh; UPS charge limit 270 kW, effective discharge limit 1,000 kW, efficiencies 0.82/0.92; TES bounds 0–1,000 kWh-th, initial energy 500 kWh-th, charge/discharge limits 300 kW-th, efficiencies 0.9/0.9; chiller input limit 400 kW and COP 5. The UPS’s nominal discharge rating is 2,700 kW, but the model restricts useful discharge to IT demand and hence to at most 1,000 kW. The time-varying IT restriction can be tighter still.

Initial temperatures ($T^{\text{IT}}, T^{\text{R}}, T^{\text{CA}}, T^{\text{HA}}$) are (28.5, 26, 20, 27)°C. Bounds at every state boundary are 18–60, 18–40, 18–23 and 18–40°C, respectively; inlet temperature is 18–30°C. Both annual cases use the same thermal bounds. A historical configuration field named `baseline_cold_aisle_setpoint_c` is not an active fixed-setpoint constraint in the retained reference model.

Auxiliary power is 53.095 kW. It was calibrated as 0.07×758.5 kW from an earlier daily IT-plus-cooling benchmark and is held constant in all retained cases. It represents approximately 7.4% of annual reference IT-plus-cooling mean demand, rather than an enforced 7% annual share. The structural grid-import limit is $1000 + 270 + 400 + 53.095 = 1723.095$ kW. No additional site-specific connection cap, ramp-rate constraint or reserve above the proportional UPS lower bound is active in these results.

S2 Workload and electricity-price inputs

S2.1 Adopted workload arrays and their interpretation

Published studies describe differing utilisation levels, workload classes and temporal flexibility across facilities [1, 3, 5, 7]. They motivate separating immediate work from work that may be rescheduled, but do not identify a universal flexible/inflexible ratio. The inputs used here are stylised profiles selected with reference to that literature. They are not claimed to be a measured site trace, a statistically representative sample of facilities or a deterministic reconstruction of a particular published trace.

The authoritative numerical inputs are `data/load_profiles.csv` and `data/shiftability_profile.csv`, whose active slots 1–96 are reproduced jointly in Table S3. The load file also contains legacy rows 97–108; annual timeline construction does not use those rows. The former file specifies flexible and inflexible arrival utilisation; the latter specifies four fractions summing to one in each active slot. These exact quarter-hour values supersede abbreviated hourly summaries when reproducing the calculations. In particular, the tranche array is used at its recorded quarter-hour resolution; no hourly table is expanded to replace it. An older literature-range table is not reproduced because its reported ranges do not consistently enclose the adopted values and it does not specify the actual solver inputs.

For an arrival in interval t , class k receives work

$$a_{t,k} = u_t^{\text{flex,arr}} \alpha_{t,k} \Delta t, \quad \ell_{t,k} = t + D_k \Delta t, \quad (D_1, D_2, D_3, D_4) = (2, 4, 8, 12). \quad (\text{S1})$$

The final execution interval starts at $\ell_{t,k}$ and closes at $\ell_{t,k} + \Delta t$. For example, a three-hour class arriving at 12:00 may execute in the interval starting at 15:00 and ending at 15:15. Thus the class names denote shifts between interval starts, not a guarantee of completion within exactly three continuous hours of an interval’s opening. Cohort creation omits quantities no larger than the 10^{-7} CPU-hour numerical tolerance.

For workload multiplier m , let $v_t = u_t^{\text{inflex}} + u_t^{\text{flex,arr}}$ in the original input. The sensitivity uses

$$u_t^{\text{flex,arr}}(m) = \min\{m u_t^{\text{flex,arr}}(1), v_t\}, \quad u_t^{\text{inflex}}(m) = v_t - u_t^{\text{flex,arr}}(m). \quad (\text{S2})$$

Tranche fractions remain fixed. The clipping rule preserves total arriving CPU-hours and avoids creating more flexible work than the total workload. A multiplier is therefore not always an exactly proportional increase in the realised flexible share.

Table S3: Actual quarter-hour workload inputs. Utilisations are fractions of rated CPU capacity; tranche shares are fractions of flexible work.

Local start	Inflexible	Flexible	α_1	α_2	α_3	α_4
00:00	0.28	0.40	0.25	0.25	0.20	0.30
00:15	0.27	0.38	0.25	0.25	0.15	0.35
00:30	0.26	0.36	0.35	0.17	0.18	0.30
00:45	0.25	0.34	0.25	0.15	0.20	0.40
01:00	0.25	0.31	0.25	0.20	0.15	0.40
01:15	0.23	0.32	0.25	0.15	0.15	0.45
01:30	0.21	0.32	0.23	0.28	0.15	0.34
01:45	0.19	0.32	0.20	0.20	0.15	0.45
02:00	0.17	0.33	0.25	0.15	0.13	0.47
02:15	0.16	0.30	0.25	0.17	0.16	0.42
02:30	0.16	0.28	0.35	0.22	0.13	0.30
02:45	0.16	0.26	0.32	0.20	0.15	0.33
03:00	0.16	0.23	0.45	0.25	0.15	0.15
03:15	0.14	0.24	0.50	0.20	0.15	0.15
03:30	0.12	0.25	0.40	0.20	0.25	0.15
03:45	0.10	0.26	0.45	0.25	0.15	0.15
04:00	0.08	0.27	0.50	0.15	0.20	0.15
04:15	0.07	0.27	0.50	0.20	0.15	0.15
04:30	0.06	0.27	0.10	0.20	0.30	0.40
04:45	0.06	0.27	0.15	0.20	0.25	0.40
05:00	0.06	0.27	0.18	0.12	0.25	0.45

Local start	Inflexible	Flexible	α_1	α_2	α_3	α_4
05:15	0.07	0.24	0.22	0.18	0.25	0.35
05:30	0.09	0.22	0.20	0.16	0.24	0.40
05:45	0.10	0.20	0.15	0.20	0.20	0.45
06:00	0.12	0.18	0.25	0.25	0.20	0.30
06:15	0.14	0.19	0.25	0.25	0.15	0.35
06:30	0.16	0.21	0.35	0.17	0.18	0.30
06:45	0.18	0.22	0.25	0.15	0.20	0.40
07:00	0.20	0.24	0.25	0.20	0.15	0.40
07:15	0.21	0.24	0.25	0.15	0.15	0.45
07:30	0.22	0.24	0.23	0.28	0.15	0.34
07:45	0.23	0.24	0.20	0.20	0.15	0.45
08:00	0.24	0.24	0.25	0.15	0.13	0.47
08:15	0.26	0.25	0.25	0.17	0.16	0.42
08:30	0.29	0.26	0.35	0.22	0.13	0.30
08:45	0.31	0.27	0.32	0.20	0.15	0.33
09:00	0.34	0.28	0.45	0.25	0.15	0.15
09:15	0.35	0.25	0.50	0.20	0.15	0.15
09:30	0.36	0.23	0.40	0.20	0.25	0.15
09:45	0.37	0.21	0.45	0.25	0.15	0.15
10:00	0.37	0.19	0.50	0.15	0.20	0.15
10:15	0.38	0.18	0.50	0.20	0.15	0.15
10:30	0.39	0.17	0.10	0.20	0.30	0.40
10:45	0.40	0.16	0.15	0.20	0.25	0.40
11:00	0.42	0.16	0.18	0.12	0.25	0.45
11:15	0.41	0.17	0.22	0.18	0.25	0.35
11:30	0.40	0.18	0.20	0.16	0.24	0.40
11:45	0.39	0.19	0.15	0.20	0.20	0.45
12:00	0.40	0.20	0.25	0.25	0.20	0.30
12:15	0.39	0.21	0.25	0.25	0.15	0.35
12:30	0.39	0.22	0.35	0.17	0.18	0.30
12:45	0.38	0.23	0.25	0.15	0.20	0.40
13:00	0.36	0.24	0.25	0.20	0.15	0.40
13:15	0.36	0.24	0.25	0.15	0.15	0.45
13:30	0.35	0.25	0.23	0.28	0.15	0.34
13:45	0.35	0.26	0.20	0.20	0.15	0.45
14:00	0.35	0.27	0.25	0.15	0.13	0.47
14:15	0.34	0.27	0.25	0.17	0.16	0.42
14:30	0.33	0.27	0.35	0.22	0.13	0.30
14:45	0.32	0.27	0.32	0.20	0.15	0.33
15:00	0.33	0.27	0.45	0.25	0.15	0.15
15:15	0.34	0.25	0.50	0.20	0.15	0.15
15:30	0.35	0.23	0.40	0.20	0.25	0.15
15:45	0.37	0.21	0.45	0.25	0.15	0.15
16:00	0.40	0.20	0.50	0.15	0.20	0.15
16:15	0.37	0.25	0.50	0.20	0.15	0.15
16:30	0.33	0.30	0.10	0.20	0.30	0.40
16:45	0.30	0.35	0.15	0.20	0.25	0.40
17:00	0.27	0.40	0.18	0.12	0.25	0.45
17:15	0.27	0.41	0.22	0.18	0.25	0.35
17:30	0.27	0.42	0.20	0.16	0.24	0.40

Local start	Inflexible	Flexible	α_1	α_2	α_3	α_4
17:45	0.26	0.43	0.15	0.20	0.20	0.45
18:00	0.26	0.45	0.25	0.25	0.20	0.30
18:15	0.26	0.45	0.25	0.25	0.15	0.35
18:30	0.26	0.45	0.35	0.17	0.18	0.30
18:45	0.26	0.45	0.25	0.15	0.20	0.40
19:00	0.26	0.45	0.25	0.20	0.15	0.40
19:15	0.26	0.45	0.25	0.15	0.15	0.45
19:30	0.25	0.46	0.23	0.28	0.15	0.34
19:45	0.25	0.46	0.20	0.20	0.15	0.45
20:00	0.25	0.47	0.25	0.15	0.13	0.47
20:15	0.26	0.46	0.25	0.17	0.16	0.42
20:30	0.27	0.45	0.35	0.22	0.13	0.30
20:45	0.28	0.44	0.32	0.20	0.15	0.33
21:00	0.29	0.41	0.45	0.25	0.15	0.15
21:15	0.29	0.41	0.50	0.20	0.15	0.15
21:30	0.29	0.41	0.40	0.20	0.25	0.15
21:45	0.30	0.41	0.45	0.25	0.15	0.15
22:00	0.30	0.40	0.50	0.15	0.20	0.15
22:15	0.27	0.40	0.50	0.20	0.15	0.15
22:30	0.25	0.41	0.10	0.20	0.30	0.40
22:45	0.22	0.41	0.15	0.20	0.25	0.40
23:00	0.21	0.42	0.18	0.12	0.25	0.45
23:15	0.21	0.42	0.22	0.18	0.25	0.35
23:30	0.21	0.42	0.20	0.16	0.24	0.40
23:45	0.21	0.42	0.15	0.20	0.20	0.45

S2.2 Prices, calendar and settlement

The source is LCCC’s IMRP actuals dataset [4], whose reference product is the Great Britain day-ahead hourly price. Source fields identify local date, chronological hourly settlement period and IMRP amount in GBP/MWh. The 2025 series covers 8,760 physical hours. Hourly prices are replicated to four consecutive quarter-hours without smoothing, giving 35,040 committed intervals. The timeline uses UTC for physical ordering and Europe/London for local-date and workload-slot labels. Spring and autumn clock changes produce 92 and 100 quarter-hours in those local days; workload values for skipped/repeated local-clock periods are correspondingly skipped/repeated. No physical UTC interval is missing or duplicated.

Negative prices remain signed in both optimisation and reported costs. There are 672 negative-price quarter-hours in the retained annual data. IMRP is a reference-price signal, not a full retail tariff. The first three hourly prices on 1 January 2026 are GBP 46.49, 40.19 and 28.99/MWh; they supply the twelve final look-ahead intervals. The exact expanded central price timeline is included as `data/price_timeline.csv`. Annual cost uses only committed intervals in calendar 2025.

S3 Complete integrated formulation

S3.1 Cohorts within a finite horizon

For horizon $\mathcal{H}_d$ beginning at s_d , combine unfinished cohorts arriving before s_d with new cohorts arriving within the horizon. A carried cohort contributes its remaining work, while a new cohort

contributes a_c ; denote either amount by $A_{c,d}$. Define

$$\mathcal{W}_{c,d} = \{t \in \mathcal{H}_d : r_c \leq t \leq \ell_c\}. \quad (\text{S3})$$

Variables $u_{c,t} \geq 0$ exist only on this set. If L_d is the last interval start in $\mathcal{H}_d$, the processing constraints are

$$\sum_{t \in \mathcal{W}_{c,d}} u_{c,t} \Delta t = A_{c,d}, \quad \ell_c \leq L_d, \quad (\text{S4})$$

$$\sum_{t \in \mathcal{W}_{c,d}} u_{c,t} \Delta t \leq A_{c,d}, \quad \ell_c > L_d, \quad (\text{S5})$$

$$u_t = u_t^{\text{inflex}} + \sum_{c:t \in \mathcal{W}_{c,d}} u_{c,t}, \quad 0 \leq u_t \leq 1. \quad (\text{S6})$$

Individual processing variables also have upper bound one. For cohorts arriving before the next core boundary, committed residual work is

$$R_{c,s_{d+1}} = A_{c,d} - \sum_{t \in \mathcal{C}_d \cap \mathcal{W}_{c,d}} u_{c,t} \Delta t. \quad (\text{S7})$$

Positive residuals above tolerance are carried with unchanged identifiers, arrival times and final eligible execution intervals. Cohorts arriving only in advisory look-ahead intervals are generated afresh on the subsequent solve; their uncommitted trial execution is not carried forward. This prevents look-ahead decisions from being mistaken for completed work.

In reference operation, all arriving work executes immediately: $u_t = u_t^{\text{inflex}} + u_t^{\text{flex,arr}}$. The same utilisation-to-power approximation and cooling equations apply. Storage actions are zero. Changing the flexible share therefore does not change total reference arrivals.

S3.2 Piecewise-linear IT power

Let $f(u) = 166.7 + (1000 - 166.7)u^{1.32}$ kW. The five breakpoints are $b_i = (i/4)^{1.5}$, $i = 0, \dots, 4$. For $b_i \leq u \leq b_{i+1}$, the implemented curve is

$$\hat{f}(u) = f(b_i) + \frac{f(b_{i+1}) - f(b_i)}{b_{i+1} - b_i}(u - b_i), \quad P_t^{\text{IT}} = \hat{f}(u_t). \quad (\text{S8})$$

The coordinated formulation imposes equality to this piecewise curve using Pyomo's DLOG representation. Equivalently, exactly one segment is active and power/utilisation are interpolated on its two endpoints; the logarithmic encoding reduces the binary representation of that selection. Reference utilisation is fixed, so its power is evaluated directly on the identical interpolation. The analytic convex curve lies at or below the chords. The retained approximation check gives a maximum error of 6.241 kW, or 0.624% of rated IT power. This is an approximation to an aggregate power relation, not a measurement-error estimate.

S3.3 UPS, meter balance and operating domains

With $z_t^{\text{U}} \in \{0, 1\}$ and all charge/discharge powers non-negative,

$$E_{t+1}^{\text{U}} = E_t^{\text{U}} + \left(\eta_{\text{U,ch}} P_t^{\text{U,ch}} - P_t^{\text{U,dis}} / \eta_{\text{U,dis}} \right) \Delta t, \quad (\text{S9})$$

$$0.5 E_{\text{nom}}^{\text{U}} \leq E_t^{\text{U}} \leq E_{\text{nom}}^{\text{U}}, \quad (\text{S10})$$

$$0 \leq P_t^{\text{U,ch}} \leq 270 z_t^{\text{U}}, \quad (\text{S11})$$

$$0 \leq P_t^{\text{U,dis}} \leq 1000(1 - z_t^{\text{U}}), \quad P_t^{\text{U,dis}} \leq P_t^{\text{IT}}, \quad (\text{S12})$$

$$P_t^{\text{grid,IT}} = P_t^{\text{IT}} - P_t^{\text{U,dis}} \geq 0, \quad (\text{S13})$$

$$P_t^{\text{grid}} = P_t^{\text{grid,IT}} + P_t^{\text{U,ch}} + P_t^{\text{C}} + P_t^{\text{T,ch}} + 53.095, \quad (\text{S14})$$

$$0 \leq P_t^{\text{grid}} \leq 1723.095. \quad (\text{S15})$$

The underlying IT-supplied grid variable also has upper bound 1,000 kW. There are no positive minimum charging/discharging powers. The battery does not export separately or supply cooling in this topology. For energy-capacity sensitivities $E_{\text{nom}}^{\text{U}} = 600m$ kWh, initial energy equals $E_{\text{nom}}^{\text{U}}$ and the reserve scales with it; power limits are unchanged. The initial central battery has 600 kWh stored but only 300 kWh between its operational bounds.

S3.4 TES and chiller

Let $z_t^{\text{T}} \in \{0, 1\}$. Chiller electrical powers are non-negative and TES thermal powers obey

$$E_{t+1}^{\text{T}} = E_t^{\text{T}} + \left(\eta_{\text{T,ch}} Q_t^{\text{T,ch}} - Q_t^{\text{T,dis}} / \eta_{\text{T,dis}} \right) \Delta t, \quad (\text{S16})$$

$$0 \leq E_t^{\text{T}} \leq E_{\text{max}}^{\text{T}}, \quad (\text{S17})$$

$$0 \leq Q_t^{\text{T,ch}} \leq 300 z_t^{\text{T}}, \quad 0 \leq Q_t^{\text{T,dis}} \leq 300(1 - z_t^{\text{T}}), \quad (\text{S18})$$

$$Q_t^{\text{T,ch}} = \text{COP} P_t^{\text{T,ch}}, \quad (\text{S19})$$

$$Q_t^{\text{cool}} = \text{COP} P_t^{\text{C}} + Q_t^{\text{T,dis}}, \quad (\text{S20})$$

$$P_t^{\text{C}} + P_t^{\text{T,ch}} \leq 400. \quad (\text{S21})$$

Central $E_{\text{max}}^{\text{T}} = 1000$ kWh-th and $E_0^{\text{T}} = 500$ kWh-th. In sensitivities capacity is $1000m$ and initial energy $500m$, with fixed power limits. UPS and TES standing losses, throughput penalties and terminal-value terms are zero in the retained central objective.

S3.5 Four-node thermal dynamics

The equations adapt the four-node representation of Cupelli et al. [2] to the modelled facility and TES. With $\Delta t_s = 900$ s and kW/kJ units, they are

$$T_t^{\text{in}} = T_{t+1}^{\text{HA}} - Q_t^{\text{cool}} / (\dot{m} c_p), \quad (\text{S22})$$

$$T_{t+1}^{\text{IT}} = T_t^{\text{IT}} + \frac{\Delta t_s}{C_{\text{IT}}} [P_t^{\text{IT}} - G_{\text{conv}}(T_{t+1}^{\text{IT}} - T_{t+1}^{\text{R}})], \quad (\text{S23})$$

$$T_{t+1}^{\text{R}} = T_t^{\text{R}} + \frac{\Delta t_s}{C_{\text{R}}} [\dot{m} \kappa c_p (T_{t+1}^{\text{CA}} - T_{t+1}^{\text{R}}) + G_{\text{conv}}(T_{t+1}^{\text{IT}} - T_{t+1}^{\text{R}})], \quad (\text{S24})$$

$$T_{t+1}^{\text{CA}} = T_t^{\text{CA}} + \frac{\Delta t_s}{C_{\text{CA}}} [\dot{m} \kappa c_p (T_t^{\text{in}} - T_{t+1}^{\text{CA}}) - G_{\text{cold}}(T_{t+1}^{\text{CA}} - T^{\text{out}})], \quad (\text{S25})$$

$$T_{t+1}^{\text{HA}} = T_t^{\text{HA}} + \frac{\Delta t_s}{C_{\text{HA}}} [\dot{m} \kappa c_p (T_{t+1}^{\text{R}} - T_{t+1}^{\text{HA}})]. \quad (\text{S26})$$

These are implicit, linear updates. Thermal boundary values are constrained variables, with opening values imposed rather than chosen freely. The hot aisle has no separate external heat-transfer term. Summing the four state equations gives the node heat balance with gross cooling multiplied by the calibrated factor κ :

$$\sum_j C_j \frac{T_{t+1}^j - T_t^j}{\Delta t_s} = P_t^{\text{IT}} - \kappa Q_t^{\text{cool}} - G_{\text{cold}}(T_{t+1}^{\text{CA}} - T^{\text{out}}). \quad (\text{S27})$$

This identity explains why gross cooling and effective cooling in the represented nodes must not be interchanged. The factor describes an aggregate calibration of flow/mixing effects; the remaining fraction is not a separately measured loss stream. The model additionally imposes

$$P_t^{\text{IT}} \leq Q_t^{\text{cool}} \leq \dot{m} c_p (T_{t+1}^{\text{HA}} - 18), \quad (\text{S28})$$

together with every temperature bound in S1. The lower inequality is an adopted restriction on using thermal accumulation for flexibility. These equations and bounds define the assessed model; validation of the original source model does not constitute validation of every adaptation here.

S3.6 Objective and event additions

Every annual horizon minimises $\sum_{t \in \mathcal{H}_d} \pi_t P_t^{\text{grid}} \Delta t / 1000$. Reference operation fixes storage powers to zero and workload to arrival, while retaining the cooling optimisation. Event trials use the same objective on their event-plus-recovery intervals and add the signed target and recovery inequalities given in the main paper. For every cohort present by recovery boundary H , residual work is its original requirement less actual pre-event and trial execution. Cohorts whose final eligible interval is within the trial must be completed; later cohorts may remain unfinished only up to their undisturbed residual plus tolerance. No aggregation of residuals across different deadlines substitutes for this per-cohort check.

S4 Annual evaluation and state transfer

Each annual case follows this procedure:

1. Build the continuous UTC price timeline and assign Europe/London date and profile-slot labels. Load the quarter-hour workload and tranche arrays.
2. Initialise UPS/TES energy and the four thermal states once, at the beginning of the year; the initial unfinished-work ledger is empty.
3. For local day d , select its 92, 96 or 100 core intervals and append twelve further intervals. Load the carried state and unfinished cohorts, and generate new arrivals throughout the horizon.
4. Solve the reference or coordinated model, and record feasibility, objective bounds and accounting checks. Both core and look-ahead contain the full facility demand.
5. Commit only core controls and processing. Accumulate signed-price settlement for those intervals, save their ending physical state and compute unfinished work from committed processing only.
6. Pass the ending state and unfinished cohorts to the next local day. Reoptimise all future controls. Repeat through 31 December.

If $\mathbf{x}$ contains the two energy states and four temperatures, physical transfer is $\mathbf{x}_{d+1}^{\text{start}} = \mathbf{x}_d^{\text{core, end}}$. There is no daily storage equality restoring the initial state. Storage and workload transfer are separate: an energy state alone cannot reconstruct which jobs remain eligible for execution.

For annual committed set $\mathcal{Y}$, reported settlement is

$$C = \sum_{t \in \mathcal{Y}} \pi_t P_t^{\text{grid}} \Delta t / 1000. \quad (\text{S29})$$

This counts every 2025 interval once. The final continuation contains ordinary January demand as well as outstanding December work, so its entire cost is not added to 2025 expenditure. The central ledger contains 0.642875 CPU-hours at midnight after 31 December, and the retained continuation serves that work within its permitted execution windows. The final continuation's feasibility does not imply that its full dispatch is committed as an additional annual day.

The implementation records each day's controls, physical states, cohort residuals, solver status and audits. Central operation also records actual processing by cohort and interval,

permitting event initialisation from the exact unfinished-work ledger. Recorded daily objectives include look-ahead decisions, whereas reported annual costs include committed decisions only. A daily MILP gap is therefore not an optimality certificate for a single full-year optimisation problem.

S5 Representative-day and flexibility-event procedures

S5.1 Day selection

Daily features from central coordinated operation are: mean price, price standard deviation, minimum and maximum price, negative-price share, mean and peak grid import, mean total CPU utilisation, daily settlement cost, and opening UPS, TES and cold-aisle states. Each feature is centred on its mean across all annual daily observations and scaled by its population standard deviation; a zero standard deviation is replaced by one. For candidate day d , the score is

$$s_d = \left[\frac{1}{12} \sum_{j=1}^{12} \left(\frac{v_{dj} - \bar{v}_j}{\sigma_j} \right)^2 \right]^{1/2}. \quad (\text{S30})$$

Candidates have 96 intervals and local date before 29 December 2025. The smallest score is selected, with date used to break a tie. This gives 22 September, score 0.13487744. The complete candidate table accompanies the package. The rule identifies a central multivariate illustration; it does not weight days by event availability, or guarantee a typical duration envelope.

S5.2 Opening state and recovery

For each hourly t_0 , load the undisturbed central trajectory and exact remaining work at that boundary. Pre-event execution is fixed by the stored history; the event solve starts at t_0 . New arrivals from that point are generated with their original class rules. Any existing cohort retains only its remaining work and remaining eligible intervals.

For duration τ , the trial ends at $H = t_0 + \tau + 3$ h. Impose a grid deviation of ΔP within ± 0.1 kW during the event, with no grid-target constraint during the recovery period. At H , require UPS/TES energy no smaller than the undisturbed states, each temperature within 0.05°C of its undisturbed value, and each cohort residual no larger than the undisturbed residual plus 10^{-7} CPU-hours. Workload eligibility, storage modes and all physical limits hold throughout. Surplus stored energy is permitted at recovery; exact equality of storage states is not required.

S5.3 Search, certification and dispatch interpretation

There are 24 start times, nine negative requests (-100 to -500 kW in 50 kW steps) and three positive requests (25, 50, 75 kW). For each pair, the assessment cap in intervals is the smaller of 48 and the number of intervals remaining in the local day. The procedure first tests that cap. If unsuccessful, it searches logarithmically for a feasible duration and checks the following interval. Standard trials have a 60 s solver limit; selected unresolved boundary trials have existing 300 s checks.

A trial with an audited feasible incumbent is evidence that its duration can be delivered under the model. A trial without an incumbent can be either proven infeasible or unresolved; these outcomes are retained separately. During the search, an unsuccessful trial can narrow the region explored even when it is unresolved. Consequently the results should be interpreted as the feasible durations found by this procedure, rather than as an exhaustive exploration of every duration.

The output flag `boundary_verified` covers both a feasible assessment cap and a proven-infeasible adjacent candidate. It is not, by itself, a proof that all longer durations are infeasible.

Since extending an event changes the recovery boundary and undisturbed terminal reference, monotonic feasibility with duration has not been established for this formulation. The main paper therefore reports certified feasible durations. It retains plus signs for the thirteen unresolved adjacent cases and explains the cap convention separately. Unmarked cells do not acquire a stronger global-maximality interpretation merely because the next tested interval was infeasible. Blank cells mean that this search certified no positive duration.

The accepted dispatch minimises signed event-plus-recovery electricity cost subject to the request. Where a solve terminates with a nonzero objective gap, it is an accepted incumbent for that objective, not a proven minimum-cost allocation. Even a proven optimum need not be unique. Thus component panels illustrate feasible coordinated mechanisms and exact electrical accounting; they are neither a unique attribution nor an independent valuation of the assets. The dashed target shown in retained plots applies during the event only, even where the graphical line extends into post-event space.

S6 Extended annual and component results

All sensitivity points in Table S4 are completed annual chains. They supersede earlier representative-week intermediate comparisons. The common reference has the same total workload arrivals as every flexible-share case; storage is disabled in that reference. Percentage-point changes are relative to the central coordinated saving, while saving percentages use the common reference cost as denominator. More decimals are retained in the accompanying CSV files for calculation.

Table S4: Full-year cost and resource sensitivity. Change is percentage points from the central coordinated saving; baseline denotes reference operation.

Case	Cost (GBP)	Saving (%)	Change (pp)	Energy (GWh)
Baseline	550,504	0.000	-5.050	6.718
Flexible workload 0.5×	531,117	3.522	-1.528	6.782
Flexible workload 0.75×	526,707	4.323	-0.727	6.787
Flexible workload 1.0×	522,703	5.050	+0.000	6.794
Flexible workload 1.25×	519,157	5.694	+0.644	6.801
Flexible workload 1.5×	516,058	6.257	+1.207	6.807
UPS capacity 0.5×	525,374	4.565	-0.485	6.771
UPS capacity 0.75×	524,001	4.814	-0.236	6.783
UPS capacity 1.25×	521,411	5.285	+0.235	6.805
UPS capacity 1.5×	520,222	5.501	+0.451	6.815
TES capacity 0.5×	524,080	4.800	-0.250	6.784
TES capacity 0.75×	523,327	4.937	-0.113	6.789
TES capacity 1.25×	522,319	5.120	+0.070	6.797
TES capacity 1.5×	522,057	5.167	+0.117	6.799

Figure S1 shows the full representative-day component schedule. It complements the event plots by showing the undisturbed electricity demands and behind-the-meter UPS discharge. The annual sensitivity values and event grid are also supplied in `data/annual_endpoints.csv` and `data/flexibility_results.csv`; those files retain the numerical values without requiring digitisation of a plot.

The annual component-cost differences are accounting terms within the same coordinated solution. In particular, electricity used to charge TES is positive in the charging-chiller row, while avoided direct cooling appears elsewhere. The UPS row is charging minus discharging evaluated at interval prices, not a storage-revenue settlement. The accounting identity remains valid under negative prices. Summed displayed values can differ from displayed totals by rounding; unrounded terms reconcile in the accompanying data.

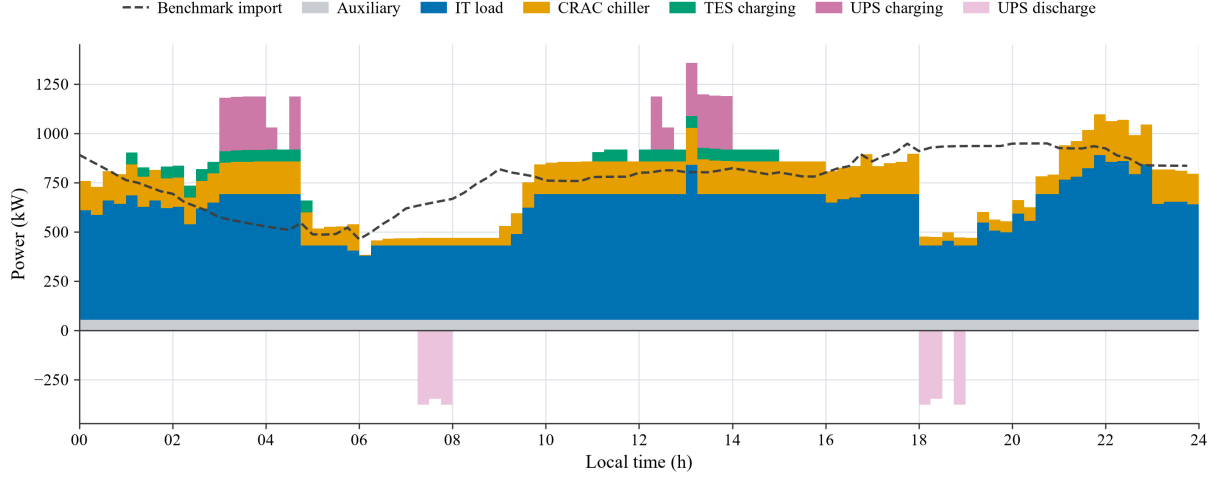


Figure S1: Undisturbed coordinated component dispatch on 22 September 2025. Positive bars are metered demands and UPS discharge offsets IT supply. The dashed line denotes workload-at-arrival reference import.

S7 Numerical checks and solution quality

The retained manuscript run record identifies Python 3.12.0, Pyomo 6.10.1 and HiGHS 1.15.1, on Windows 11 with an eight-core AMD Ryzen 7 9800X3D. Annual scenarios request a relative MILP gap of 0.1%, with a standard daily time limit of 300 s and a normal acceptance threshold of 1%. Feasible exceptions in sensitivity cases are disclosed below rather than treating every solve as proven optimal. The main paper’s 5.050% is computed from the retained annual settlements, not by summing daily objective bounds.

All fourteen annual summaries contain 365 local dates and 35,040 committed intervals. Their stored checks cover timestamp continuity, physical state handoff, workload conservation, storage mode exclusivity and signed-price settlement reconciliation. The stated audit tolerances are 10^{-5} in the native state units, 10^{-3} kW for power flows and 10^{-7} CPU-hours for work. Central operation also stores a cohort execution trace. The two central time-limited daily solutions, on 25 May and 4 October, have reported relative gaps of approximately 0.605% and 0.385%, within the normal threshold.

Table S5: Original annual-chain solver records. Non-optimal denotes the stored termination classification; diagnostic reruns are reported separately.

Case	Non-optimal horizons	Maximum recorded gap (%)
Baseline	0	0.000
Flexible workload 0.5×	3	8.201
Flexible workload 0.75×	1	1.074
Flexible workload 1.0×	2	0.605
Flexible workload 1.25×	3	2.032
Flexible workload 1.5×	2	1.181
UPS capacity 0.5×	2	2.315
UPS capacity 0.75×	2	2.441
UPS capacity 1.25×	2	0.846
UPS capacity 1.5×	2	0.870
TES capacity 0.5×	2	0.928
TES capacity 0.75×	2	1.478
TES capacity 1.25×	2	0.601
TES capacity 1.5×	2	0.996

Table S5 records the maximum original reported gap for each retained annual chain, including values above the normal threshold. Existing 900 s exact-state reruns of selected 25 May horizons provide additional evidence: the 0.75 UPS and TES cases achieve gaps of 0.627% and 0.959%; the 0.75 and 1.25 workload cases retain gaps of 1.015% and 2.032%. Across these intermediate checks, the largest committed-day cost change is GBP 0.281, with unchanged closing workload and physical-state differences no greater than 2.14×10^{-14} . The 0.5 workload case retains a 5.652% gap in its existing 900 s check, but its reported absolute primal-bound width is GBP 0.102 and committed cost changes by approximately GBP 0.002. Near-zero negative objectives can produce large relative gaps despite small absolute bounds.

These reruns are diagnostics of particular horizons, not new full-year scenarios substituted into the table. Their limited effects support the displayed result precision in those checks; they do not prove a full-year global optimum or retrospectively certify every recorded exception. Original and diagnostic gap values must not be mixed into a single claim that all annual solves meet 1%.

The existing year-end assessment confirms service of the 0.642875 CPU-hours outstanding at midnight. Its complete final look-ahead contains 2,313.104 kWh and GBP 89.542, including ordinary January demand. Existing terminal-value tests, using median-to-maximum observed 1 January prices with conversion adjustments, change annual cost by at most 0.000381% in the tested earlier central chain. That report predates the cohort-trace central result and quotes a 5.049% saving. It is supporting evidence for the terminal convention, not a replacement for the current 5.050% result or a separate terminal-sensitivity proof for every capacity case.

For the event study, the retained summary records 288 cells, of which 275 carry the boundary/cap flag and thirteen an unresolved-next-step flag. Positive accepted dispatches meet the event solver-gap criterion of 1% and the physical audits. Maximum target error is 0.1 kW, terminal-temperature error 0.05°C , excess cohort residual 10^{-7} CPU-hours and component-sum residual approximately 4.18×10^{-12} kW. These checks establish consistency with the formulated constraints. They do not eliminate the search qualifications in S5, model approximation or uncertainty in the adopted facility parameters.

S8 UPS investment-screen calculations

Let the additional annual electricity-cost saving for energy capacity E be $\Delta S(E) = C^{\text{op}}(600) - C^{\text{op}}(E)$. The retained results give GBP 1,292.2794/year for $E = 750$ kWh and GBP 2,481.4101/year for $E = 900$ kWh. Relative to the central capacity, the additional nameplate energies are 150 and 300 kWh, but the additional energy between operational bounds is 75 and 150 kWh because the 50% reserve fraction remains fixed.

For constant real annual benefits, additional investment I and additional annual non-electricity costs O , the screen is

$$\Delta S - O - I \text{CRF}(r, n) \geq 0, \quad \text{CRF}(r, n) = \frac{r(1+r)^n}{(1+r)^n - 1}. \quad (\text{S31})$$

At $r = 0$, CRF is $1/n$. The capital recovery factor converts initial expenditure into an equivalent annual amount [6]. Setting $O = 0$ gives $I_{\text{BE}} = \Delta S/\text{CRF}$. Dividing by added *nameplate* kWh gives a threshold directly comparable with a cost expressed on that same basis. The ten-year, 8% case has CRF 0.14902949. Table S6 shows additional illustrative rate/life combinations; these are assumptions for interpreting the result, not empirical estimates of project finance or battery life.

No battery market price is assumed, so the calculation establishes a maximum supported expenditure rather than a positive investment recommendation. Comparing a vendor quotation would require its currency, price year, chemistry, installed scope, integration work, reserve policy and usable/nameplate capacity convention. Existing power electronics are held fixed in the

Table S6: Illustrative supported investment per added nameplate kWh, in GBP/kWh, before additional non-electricity costs.

Life (years)	Real rate (%)	CRF	750 kWh case	900 kWh case
5	4	0.22463	38.35	36.82
5	8	0.25046	34.40	33.03
5	12	0.27741	31.06	29.82
10	4	0.12329	69.88	67.09
10	8	0.14903	57.81	55.50
10	12	0.17698	48.68	46.74
15	4	0.08994	95.79	91.96
15	8	0.11683	73.74	70.80
15	12	0.14682	58.68	56.34

dispatch sensitivity; an upgrade requiring additional converter capacity cannot be evaluated by changing only a battery price per kWh.

Losses already enter the dispatch costs through efficiency and must not be charged twice. Degradation, additional O&M, replacements and financing/tax detail are omitted. Introducing additional costs reduces supported investment under the same annual-benefit assumption. A more complete lifetime treatment could also change the benefit itself. In particular, initial UPS energy is proportional to capacity and terminal energy has no credit. The retained annual cases finish with 300, 375 and 450 kWh for 600, 750 and 900 kWh capacity, respectively. The resulting single-year differences are not independently established steady-state annual benefits. The earlier central terminal check in S7 does not resolve this for all enlarged-capacity cases.

Accordingly, repeating the 2025 incremental saving over n years is an explicit conditional assumption. The thresholds are gross screens before omitted costs, not forecasts of a site's investment return. They exclude the possible value of greater backup provision and any revenues from grid services. Neither the event heatmap nor the annual component accounting supplies those revenues. If credible installed incremental costs exceed the stated threshold, the assumed energy-arbitrage saving alone is insufficient; if they are lower, the screen is favourable only within the stated operational and financial assumptions.

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